higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

FLUID MECHANICS N5

(8190205)

26 November 2019 (X-Paper)
09:00–12:00

Nonprogrammable calculators and drawing instruments may be used.

This question paper consists of 5 pages and a formula sheet of 2 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
FLUID MECHANICS N5
TIME: 3 HOURS
MARKS: 100

NOTE: If you answer more than the required number of questions only the required number will be marked. Clearly cross out ALL work you do NOT want to be marked.

INSTRUCTIONS AND INFORMATION

1. Answer any FIVE questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. $g = 9{,}81\ m/s^2$.
5. Show ALL units in the answers.
6. Write neatly and legibly.

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QUESTION 1

1.1 Explain the purpose of bearing lubrication in a mechanism and give TWO advantages thereof. (2 + 2) (4)

1.2 Define the *absolute coefficient of viscosity of a fluid* and relate kinematic viscosity to this value (include units). (1 + 2) (3)

1.3 A 150 mm diameter shaft rotates at a rotational frequency of 1 200 r/min inside a 150,3 mm diameter bearing which is 170 mm long. The oil lubricating this bearing has an absolute coefficient of viscosity of 0,02 Pa.s with a density of 830 kg/m^3.

Calculate:

1.3.1 The power loss due to viscous forces (8)

1.3.2 The kinetic viscosity of the oil (2)

1.3.3 The power loss due to viscous forces if the shaft is not rotating, but is moving axially through the bearing with a velocity of 5 m/s (3)

[20]

QUESTION 2

2.1 Explain Pascal's law for a closed hydraulic system (brake system) in relation to a force exerted on a unit area. (2)

2.2 What is the function of an accumulator in a hydraulic system? (1)

2.3 A brake system consisting of a master cylinder with a diameter of 35 mm is connected to a slave cylinder with a 100 mm diameter by means of a pipe. An external force of 800 N is applied onto the master-cylinder piston to activate the brake system. The working fluid in the system has an isothermal bulk modulus of 1,8 GPa and the average Young's modulus of elasticity for the hydraulic brake system is 50 GPa. The volume of the fluid in the master cylinder and the connecting pipe is twice the volume of the total fluid stroke length of 20 mm.

Calculate:

2.3.1 The force exerted by the slave-cylinder piston side (3)

2.3.2 The play on the master-cylinder piston if the slave-cylinder piston is applying a full force at the end of its stroke (assume $K_{cylinder} = E/2,5$) (14)

[20]

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QUESTION 3

A drum with an internal diameter of 1,45 m and 2 m long is filled with an oil with a relative density of 0,9. The total mass of the empty drum is 66 kg.

Calculate:

3.1 The internal force the oil exerts on the curved side of the drum and the centre of pressure position of this force if the drum stands on one of the circular sides (6)

3.2 The height of the column of water required to produce this pressure (5)

3.3 The centre of pressure position of the force in QUESTION 3.2 (2)

3.4 The draught of the drum (if now filled with oil) if it is floating with its longest side vertically in seawater (use 1 025 kg/m^3 for density) (7)

[20]

QUESTION 4

4.1 Which criteria are used to determine whether flow lamina or turbulent exists in a fluid-flowing field? Write down the formula (equation) and value that can be used to determine the type flow. (6)

4.2 Briefly state Bernoulli's theorem in relation to a fluid-flowing system. (2)

4.3 A jet of water sprays vertically upwards from a nozzle at a velocity of 20 m/s. The diameter of the jet of water at the exit to the nozzle is 35 mm.

Calculate:

4.3.1 The velocity of the jet 2,5 m above the base of the jet (assuming the jet remains circular and has no friction) (4)

4.3.2 The diameter of the jet at the point 2,5 m above the base (3)

4.3.3 The power of the jet (5)

[20]

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QUESTION 5

5.1 Briefly explain how a pitot tube is used to measure the velocity in a fluid-flowing system. Write down the equation used to determine its function. (4)

5.2 A pitot tube is used as a flow-velocity measuring tool in a 200 mm diameter pipeline system with flowing water. The pressure difference between the pitot-tube orifice and the static-orifice energy point is indicated in a mercury manometer head as 33,8 mm of mercury rise. The legs of the mercury manometer are filled with water. The average velocity of flow in the system is 0,82 times the central velocity of flow.

Calculate the rate of flow in the water pipeline (assume the coefficient of the tube as unity). (8)

5.3 A 40 mm diameter pipe is suddenly reduced to 15 mm in diameter. A valve with a shock factor (k) of 2,55 is fitted in the 15 mm diameter pipe. The flow rate in the system is 2 ℓ/s.

Calculate:

5.3.1 The flow velocities in the pipes (2)

5.3.2 The total head loss due to shock in the components if $C_c = 0,65$ (6)

[20]

QUESTION 6

The vanes of a water turbine are mounted on a turbine wheel at a mean diameter of 1,7 m and the wheel rotates at 168,517 r/min. The water nozzle is mounted at 19° to the direction of rotation of the wheel and the velocity of a stream of water is 40 m/s at a mass rate of 28,5 kg/s discharged through the nozzle outlet. A loss of 12% occurs in the relative velocity due to the water friction over the blades. The vanes and nozzle are arranged in such a way that no axial thrust occurs.

6.1 Determine the blade velocity. (1)

6.2 Construct, to a scale of 5 mm:1 m/s, a vector diagram from the given information. Insert ALL flow velocities (m/s) and angles in the correct positions on the vector diagram. Marks will be deducted if instructions are NOT followed. (14)

6.3 Determine the theoretical power output of the turbine according to the diagram. (3)

6.4 Determine the efficiency of the turbine according to the diagram. (2)

[20]

TOTAL: 100

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FORMULA SHEET

1. $\rho = \frac{m}{v}$

2. $SG = Rel = \frac{\rho_{\text{substance}}}{\rho_{water}}$

3. $\text{Specific}\,\omega = \frac{\text{weight}}{\text{volume}} = \rho g$

4. $P = \frac{F}{A}$

5. $P_{\text{absolute}} = P_{\text{gauge}} + P_{\text{atmospheric}}$

6. $P_{\text{gauge}} = \rho g h$

7. $F_{\text{Surface tension}} = \sigma 2\pi R$

8. $\Delta P = P_i - P_o = \frac{2\sigma}{R} = \frac{4\sigma}{D}$

9. $F_{\text{viscous}} = \frac{\mu A v}{t}$ and $\upsilon = \frac{\mu}{\rho}$

10. $K_e = \frac{P}{\varepsilon_v}$

11. $\varepsilon_v = \frac{\Delta V}{V}$

12. $\frac{1}{K_e} = \frac{1}{K_\ell} + \frac{1}{K_c} + \frac{V_g}{V_t}\left(\frac{1}{K_g}\right)$

13. $K_g = \delta P$ and $K_c = \frac{E}{2{,}5}$

14. $F_{\text{hydrostatic}} = \rho g A \bar{y}$

15. $\bar{h} = \frac{I_g Sin^2\theta}{A\bar{y}} + \bar{y}$

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16. $I_{g(\text{rectangular})} = \dfrac{bd^3}{12}$

17. $I_{g(\text{circular})} = \dfrac{\pi D^4}{64}$

18. $W = R = \rho g V$

19. Q or $\overset{\circ}{V} = A_1 u_1 = A_2 u_2$; $\overset{\circ}{m} = \rho \overset{\circ}{V}$; $\overset{\circ}{W} = g\overset{\circ}{m} = \rho g A u$; $P = H\overset{\circ}{W} = \rho g Q H$

20. $\dfrac{P_1}{\rho g} + \dfrac{u_1^2}{2g} + Z_1 + \dfrac{P_{\text{pump}}}{\overset{\circ}{W}} = H_{\text{total}} = \dfrac{P_2}{\rho g} + \dfrac{u_2^2}{2g} + Z_2 + \dfrac{P_{\text{motor}}}{\overset{\circ}{W}} + \dfrac{P_{\text{turbine}}}{\overset{\circ}{W}} + h_{\text{loss}}\ (J/N, m)$

21. $\dfrac{P_{\text{turbine}}}{\overset{\circ}{W}} = \text{Turbine head}$; $\dfrac{P_{\text{pump}}}{\overset{\circ}{W}} = \text{Pump head}$; $\eta = \dfrac{P_F}{P_m} \times 100$; $R_e = \dfrac{\rho v D}{\mu}$

22. $\underline{h_{\text{loss}}\ (J/N) \text{ or } m}$:

23. $h_s = k\,\dfrac{u^2}{2g}$; $h_s = \left(\dfrac{1}{C_c} - 1\right)^2 \dfrac{u^2}{2g}$; $h_a = h(1 - C^2{}_v)$; $h_f = 4f\left(\dfrac{L_e}{d}\right)_T \dfrac{u^2}{2g}$

24. $h_s = \dfrac{(u_1 - u_2)^2}{2g}$

25. $F_{\text{inlet}} = \overset{\circ}{m} u_1 + P_1 A_1$ and $F_{\text{exit}} = \overset{\circ}{m} u_2 + P_2 A_2$

26. Flat plate: Stationary $F = \rho A u^2$ Moving $F = \rho A(u - u_m)^2$ Angle $F = \rho A u^2 Cos\theta$

27. Curved: X – direction $F_x = \rho A u^2 (1 + Cos\theta)$ Y – direction $F_y = \rho A u^2 Sin\theta$

28. $U_m = \dfrac{\pi D n}{60}$; $P = \overset{\circ}{m} V_{w_t} u_m$; $\eta = \dfrac{2V_w u_m}{u_1^2} \times 100$

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